

SuperNeo NuMetal

One-page repository summary - generated from local repo evidence only

What it is

SuperNeo NuMetal is a macOS 14+ Swift library implementing the SuperNeo lattice folding protocol over the Goldilocks field, with CCS shapes, sum-check proofs, Ajtai commitments, proof envelopes, and optional Metal kernels.

What it's for

It is for building, testing, and benchmarking post-quantum folding-proof primitives and verifier reductions around CCS instance-witness pairs. The repo includes protocol tests, CPU/Metal differential checks, and benchmark profiles for prover, verifier, envelope, and kernel costs.

Could it be used in systems? What kind?

- **Yes, as a library/protocol component** in systems that need recursive proof composition or folding, not as a standalone deployed service.
- Repo/paper evidence names incrementally verifiable computation (IVC), proof-carrying data (PCD), recursive SNARK/proof compression, and post-quantum signature aggregation as relevant system contexts.
- Operational deployment examples, API server integration, persistence, networking, and production readiness guidance: **Not found in repo.**

Repo evidence used

- README.md and Docs/Benchmarking.md for package purpose, targets, tests, benchmark groups, and Metal baselines.
- Package.swift and SuperNeo-NuMetal/ source files for library target, public API, CPU/Metal backends, protocol flow, and serialization.
- SuperNeo-NuMetalTests/README.md and XCTest names for correctness coverage across algebra, commitments, protocol, CE opening, envelopes, and Metal.

What it does

- Models CCS shapes, sparse matrices, public inputs, witnesses, evaluation claims, and canonical digests.
- Implements Goldilocks field arithmetic, degree-2 extension elements, cyclotomic ring elements, and SuperNeo embeddings.
- Builds Ajtai commitment keys and linearly homomorphic commitments for packed witnesses.
- Constructs fold proofs through sum-check, PiCCS claims, random linear combination (PiRLC), and decomposition (PiDEC).
- Verifies fold reductions, terminal CE relation checks, terminal fold proofs, and compressed/public proof envelopes.
- Provides CPU reference paths plus optional Metal acceleration for field/ring kernels, Ajtai commits, and transformed evaluations.
- Includes XCTest slices and benchmark scripts that gate correctness and measure folds, reductions, terminal verification, envelopes, stages, and kernels.

How it works

Inputs CCSShape + instances, witnesses, prior claims, Ajtai key	Prover SuperNeoProver builds transcript-bound sum-check, PiCCS, PiRLC, PiDEC
Execution CPU reference backend or MetalExecutionContext/SuperNeoMetalWorkspace for accelerated commits + evaluations	Proof artifacts FoldProof, TerminalFoldProof, CEOpeningProof, and versioned proof envelopes
Verifier SuperNeoVerifier reduces fold, checks output claims, validates terminal CE openings	Data binding SHA-256 digests bind shapes, statements, transcript domains, headers, and envelope bodies

Key constraints surfaced by repo

- Swift package product: library target **SuperNeo_NuMetal**; platform: macOS 14+; no external SwiftPM dependencies.
- Metal is optional: benchmarks and workspace paths use Metal when a device/context is available; CPU paths remain the correctness baseline.
- End-user UI, CLI product, persistent storage model, and network/API boundary: **Not found in repo.**